

# BUYU DENG

School of the Gifted Young, University of Science and Technology of China  
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## EDUCATION

### University of Science and Technology of China

B.S. Candidate in Geophysics, School of the Gifted Young  
GPA: 3.62/4.30; rank: 3/20

Selected Coursework: Programming in Geophysics (94), Intro. to Seismic Exploration (98), Geophysical Data Processing and Analysis (93), Linear Algebra (90), Probability & Statistics (92), Theoretical Mechanics (95), Python & Deep Learning (A).

Hefei, China

Expected June 2027

## RESEARCH INTERESTS

AI for exploration geophysics; seismic interpretation, geomodeling, rock physics, and forward modeling.

## RESEARCH EXPERIENCE

### Stanford Center for Earth Resources Forecasting (SCERF), Stanford University

Undergraduate Research Intern; advisor: Prof. Tapan Mukerji

Stanford, CA

July – October 2026

#### *CO<sub>2</sub> Storage Modeling and Rock Physics Agent Development*

- Compared five vertical grid resolutions (0.04–0.92 ft) in ECLIPSE 300 CO<sub>2</sub>STORE simulations using fixed-time plume geometry, fixed-location saturation histories, space-time plume metrics, and computational cost; selected 0.12 ft for subsequent modeling.
- Built a facies-based rock physics workflow using mineral averaging, Hertz–Mindlin dry frames, Wood fluid mixing, and Gassmann substitution to map geology and CO<sub>2</sub> saturation to elastic properties.
- Generated baseline and time-lapse seismic responses for 20 years of CO<sub>2</sub> injection and 80 years after shut-in; interpreted horizon-relative amplitude slices in Petrel.
- Designed an agent workflow around the Stanford Rock Physics and Borehole Geophysics (SRB) model library exposed through Model Context Protocol (MCP) tools for model discovery, input validation, and traceable calculations.
- Conducted 45 independent agent runs across Smeaheia, glass-bead, and White patchy-saturation benchmarks under three resource settings: no references, textbook guidance only, and textbook guidance plus SRB tools.
- Without references, agents made unsupported model assumptions; textbook guidance matched reference calculations. SRB MCP tools enabled direct model calls and matched the references when used correctly, but some runs misinterpreted tool outputs and produced incorrect predictions.

Presentation: [Final slides \(PDF\)](#)

### Computational Interpretation Group (CIG), University of Science and Technology of China

Undergraduate Researcher; advisor: Prof. Xinming Wu

Hefei, China

August 2024 – Present

#### *Geo-VAE for 3D Geophysical Data Compression and Latent Processing*

- Adapted Wan-VAE with cached causal pseudo-temporal resampling along the crossline axis, achieving isotropic  $8 \times 8 \times 8$  downsampling and 31.0:1 scalar compression for  $256^3$  cubes.
- Demonstrated strong reconstruction fidelity across five 3D geophysical modalities while achieving a favorable balance of GPU memory use and inference time among tested state-of-the-art video VAEs on  $128^3$ – $512^3$  cubes.
- Built a unified compact latent space for RGT-conditioned seismic generation and complete-volume  $768^3$  denoising, enabling condition-responsive generation and scalable field-volume processing.

Code: [github.com/BuyuDeng/Geo-VAE](https://github.com/BuyuDeng/Geo-VAE)

## RESEARCH EXPERIENCE (CONTINUED)

### *3D All-in-One Seismic Denoising*

- Built a synthetic 3D seismic dataset with random noise, acquisition footprints, and migration artifacts; trained an all-in-one denoising network based on Degradation Representation Learning.

Code: [github.com/BuyuDeng/NDR\\_Seismic](https://github.com/BuyuDeng/NDR_Seismic) Noise addition slides: [Download PDF](#)

## ADDITIONAL RESEARCH PROJECTS

### *PHY-to-BGC Ocean Biogeochemical Emulation*

- Built a 2.5D full-image U-Net emulator mapping global physical ocean fields to coupled biogeochemical variables, including evaluation under anomalous ocean-climate conditions. Manuscript in preparation.

### *GNSS–NISAR Ionospheric Super-Resolution Reconstruction*

- Aligned GNSS TEC-derived phase with NISAR split-spectrum interferometric phase; tested a residual U-Net on 28 paired samples and found that sparse GNSS input preserved broad trends but did not recover NISAR-scale structure.

Report: [English project report \(PDF\)](#)

## PUBLICATIONS AND PRESENTATIONS

**B. Deng**, Y. Dou, Z. Guo, and X. Wu. "Geo-VAE: A Unified Variational Autoencoder for 3D Geophysical Data Compression and Latent Processing." Submitted to *Geophysics*; major revision requested, 2026.

**B. Deng**. "Geo-VAE: A Unified Variational Autoencoder for 3D Geophysical Data Compression and Latent Processing." Oral presentation, International Meeting for Applied Geoscience & Energy (IMAGE 2026), Houston, TX, August 2026. Paper (PDF)

**B. Deng**. "CO<sub>2</sub> Storage Modeling and Rock Physics Agent Development." SCERF Seminar, Stanford University, September 2026. [Download slides \(PDF\)](#)

Y. Dou, X. Wu, Z. Guo, H. Gao, and **B. Deng**. "Implicit Structural Modeling via Generative Diffusion Frameworks." *arXiv:2606.07165*, 2026. [arxiv.org/abs/2606.07165](https://arxiv.org/abs/2606.07165).

## RESEARCH SKILLS

Python, seismic interpretation, forward modeling, rock physics, CO<sub>2</sub> flow simulation, ECLIPSE 300, and Petrel.

## HONORS AND AWARDS

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|-----------------------------------------------------------------------------|------|
| Zhao Jiuzhang Elite Program Scholarship, School of Earth and Space Sciences | 2025 |
| Rose Fund Endeavor Scholarship, School of the Gifted Young                  | 2024 |
| Zhao Jiuzhang Elite Program Scholarship, School of Earth and Space Sciences | 2024 |
| Freshman Scholarship, University of Science and Technology of China         | 2023 |

## FIELD EXPERIENCE AND LEADERSHIP

**IPGP Geophysical Field Camp**, Institut de Physique du Globe de Paris August 2025  
Conducted geophysical surveys and joined academic seminars with IPGP faculty.

**Head (Minister)**, Department of Cultural & Sports, College Student Union June 2025 – January 2026